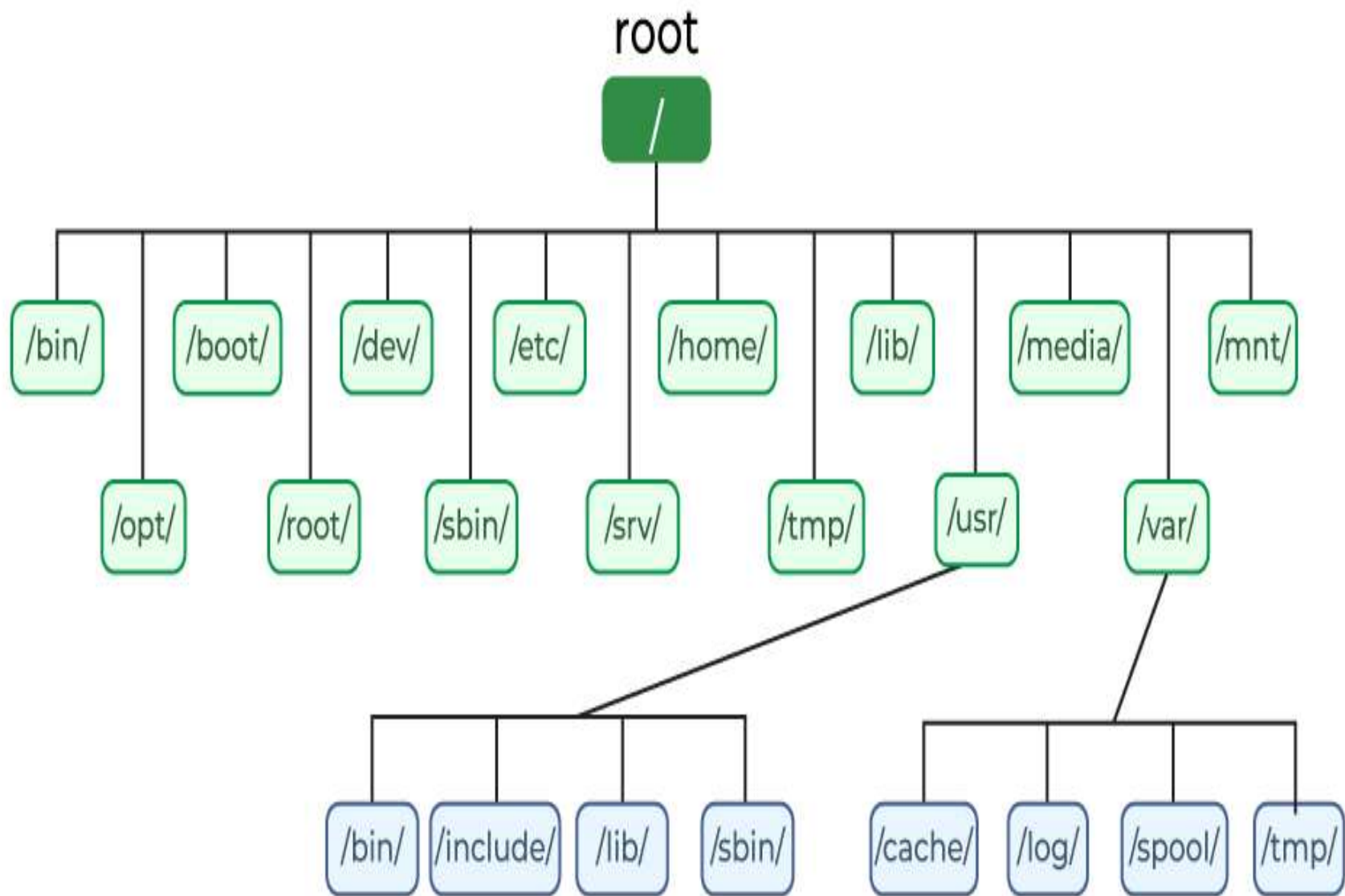


Unix File Management

Unix File Management

- * Unix file management centers around a hierarchical file system, organized like an inverted tree with the "root" directory (/) at the top. This system stores all data, including user files, system configurations, and even hardware devices, as files within directories.
- * **UNIX**, everything is treated as a **file** - whether it's text, a program, a device, or even a directory. File management is the process of **creating, storing, organizing, accessing, and controlling files** in the system.
- * All files are managed in a **hierarchical directory structure** starting from the **root(/)**.



File Types

- * Unix **Linux/Unix**, files are treated as *objects* and almost **everything is considered a file** (including devices and processes).
- * The OS classifies files into different **types**, which you can check using commands like `ls-l` or `file`.
- * **File type in linux.**
- * **Regular file (-)** → Contains normal data (text, binary, images, programs). Ex. `File.txt`, `photo.jpg`
- * **Directory (d)** → Special file that holds references (entries) to other files or directories. Ex. `/home`, `/`
- * **Character Device File (c)** → Represents hardware devices that handle data **character by character**.
Ex: `/bca/sem5`, `/bca/os`

File Types

- * **Block Device File (b)** → Represents hardware devices that handle data in **blocks**. Ex. `/bca/b1`
- * **Named Pipe (FIFO) (P)** → Special file for **inter-process communication (IPC)**.
- * Data written by one process can be read by another in a first-in-first-out manner. Ex. Created using `mkfifo`
- * **Socket(s)** → Used for communication between processes, even across networks. Ex `./var/run/a.sock`
- * **Symbolic Link (l)** → A pointer or shortcut to another file (like Windows shortcuts). Ex. `Ln-s/etc/passwd link_passwd`

File Permissions

- * Linux is a **multi-user operating system**, so permissions are critical to control **who can read, write, or execute** files.
- * Every file/directory has **three categories of users**.
- * **Owner (u)** → the user who created the file.
- * **Group (g)** → a set of users belonging to the file's group.
- * **Others (o)** → everyone else on the system.
- * **r** → Read (view file content).
- * **w** → Add/remove files inside the directory
- * **X** → Enter the directory (cd) and access files inside.

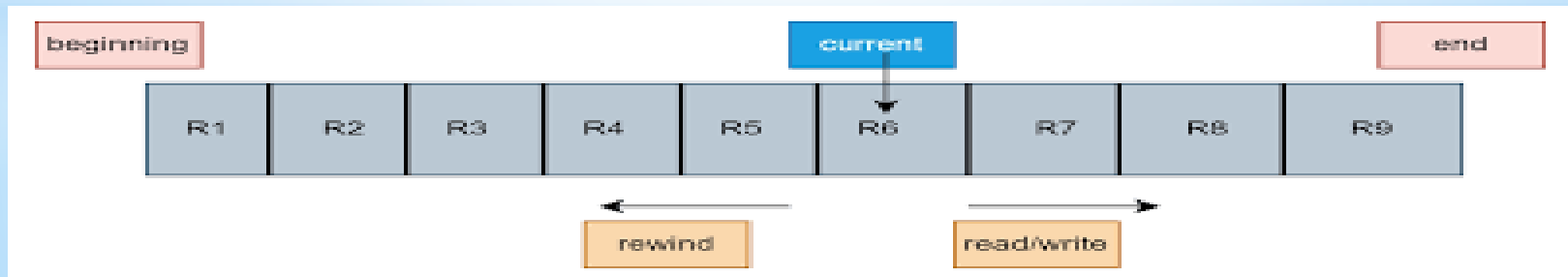
Access Method In Operating System

What are File Access Method

- * A file is a collection of bits/bytes or lines that is stored on secondary storage devices like a hard drive.
 - * File access methods in OS are nothing but techniques to read data from the system's memory.
 - * When a file is used, information is read and accessed into computer memory and there are several ways to access this information of the file.
 - * There are three ways to access a file into a computer system.
- 1) Sequential-Access
 - 2) Direct Access
 - 3) Index Sequential Method.

1). Sequential-Access

- * It is the simplest access method. Information in the file is processed in order, one record after the other.
- * This mode of access is by far the most common; for example, editor and compiler usually access the file in this fashion. Read and write make up the bulk of the operation on a file.
- * A read operation -*read next*- read the next position of the file and automatically advance a file pointer, which keeps track I/O location.
- * Similarly, for the -*write next*- append to the end of the file and advance to the newly written material.



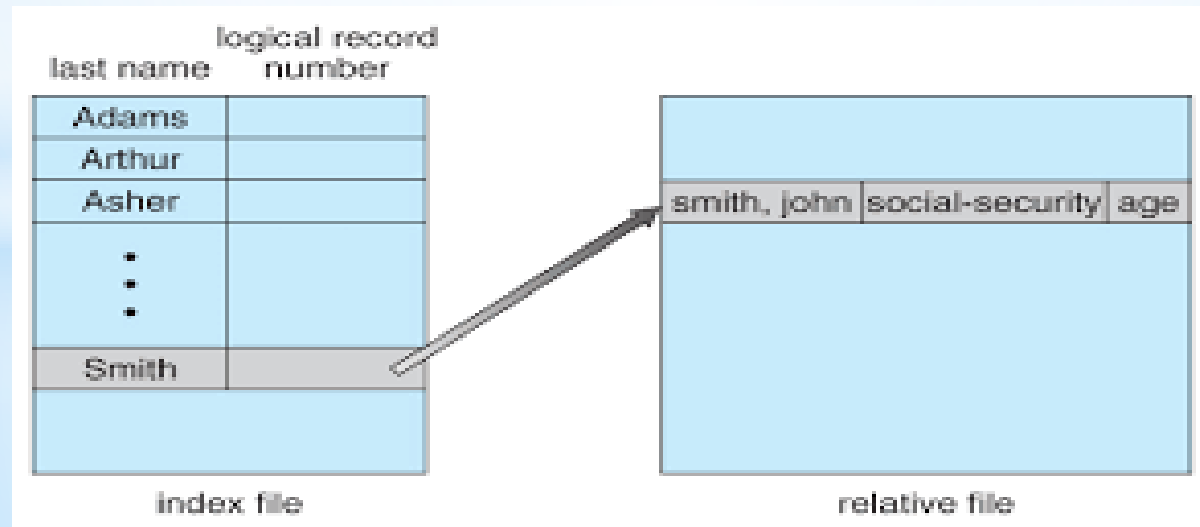
2). Direct-Access Method

- * Another method is *direct access method* also known as *relative access method*. A fixed-length logical record that allows the program to read and write record rapidly. in no particular order.
- * The direct access is based on the disk model of a file since disk allows random access to any file block. For direct access, the file is viewed as a numbered sequence of block or record.
- * There is no restriction on the order of reading and writing for a direct access file.
- * A block number provided by the user to the operating system is normally a *relative block number*, the first relative block of the file is 0 and then 1 and so on.



3). Index sequential Method

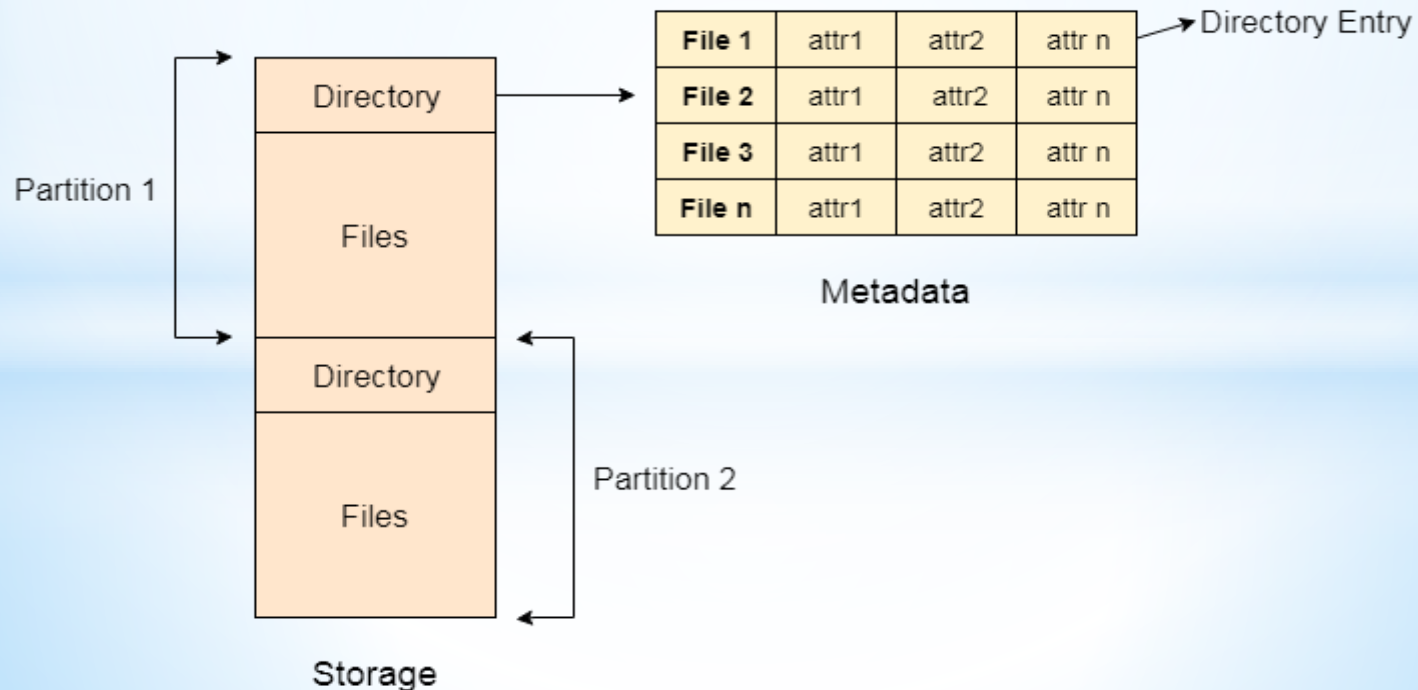
- * It is the other method of accessing a file that is built on the top of the sequential access method.
- * These methods construct an index for the file. The index, like an index in the back of a book, contains the pointer to the various blocks.
- * To find a record in the file, we first search the index, and then by the help of pointer we access the file directly.



Directory Structure in Operating System

What is a directory?

- * A Directory is a container that is used to contain folders and file. It organizes files and folders into a hierarchical manner.
- * To get the benefit of different file systems on the different operating systems, A hard disk can be divided into the number of partitions of different sizes. The partitions are also called volumes or mini disks.
- * Each partition must have at least one directory in which, all the files of the partition can be listed. A directory entry is maintained for each file in the directory which stores all the information related to that file.

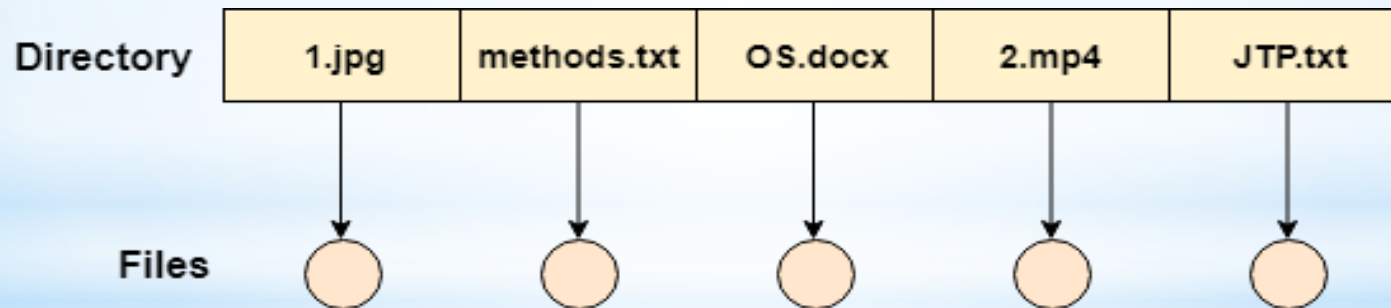


*A Directory can be support five part in following.

- 1) Single Level Directory
- 2) Two level Directory
- 3) Tree Level Directory
- 4) Acyclic- Graph Directories
- 5) General Graph Directory

1) Single Level Directory

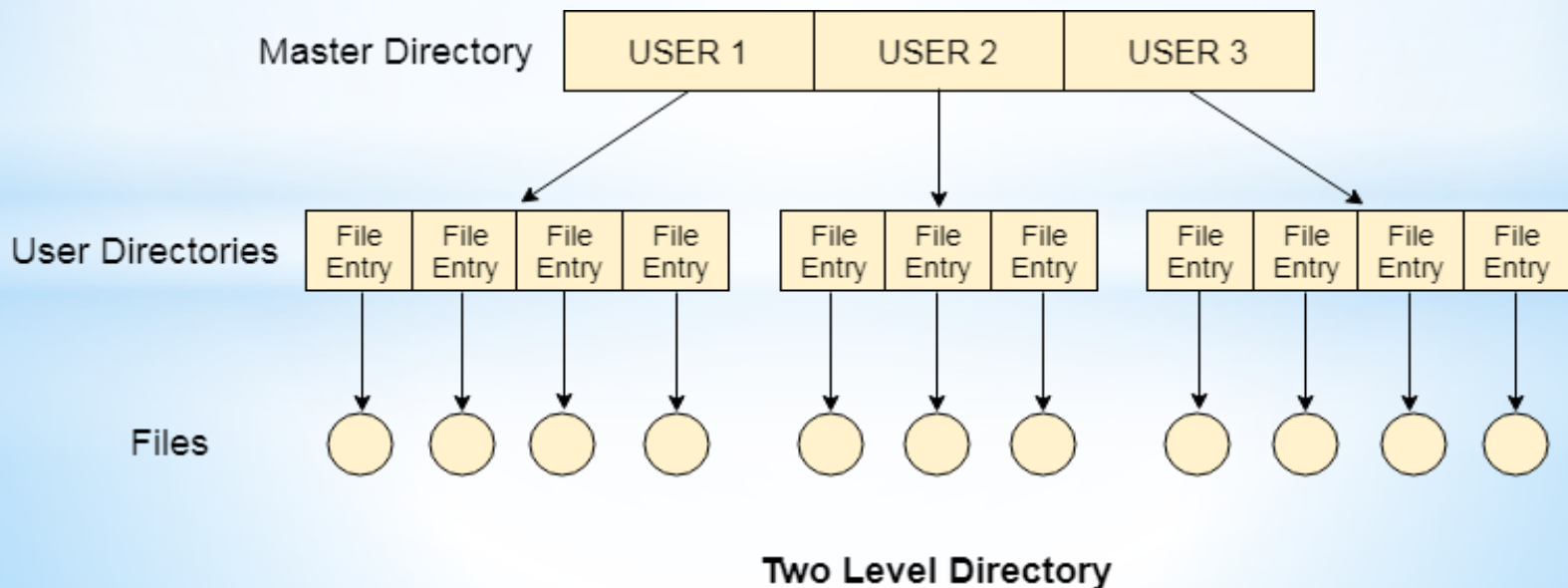
- *The simplest method is to have one big list of all the files on the disk.
- *The entire system will contain only one directory which is supposed to mention all the files present in the file system.
- *The directory contains one entry per each file present on the file system.



Single Level Directory

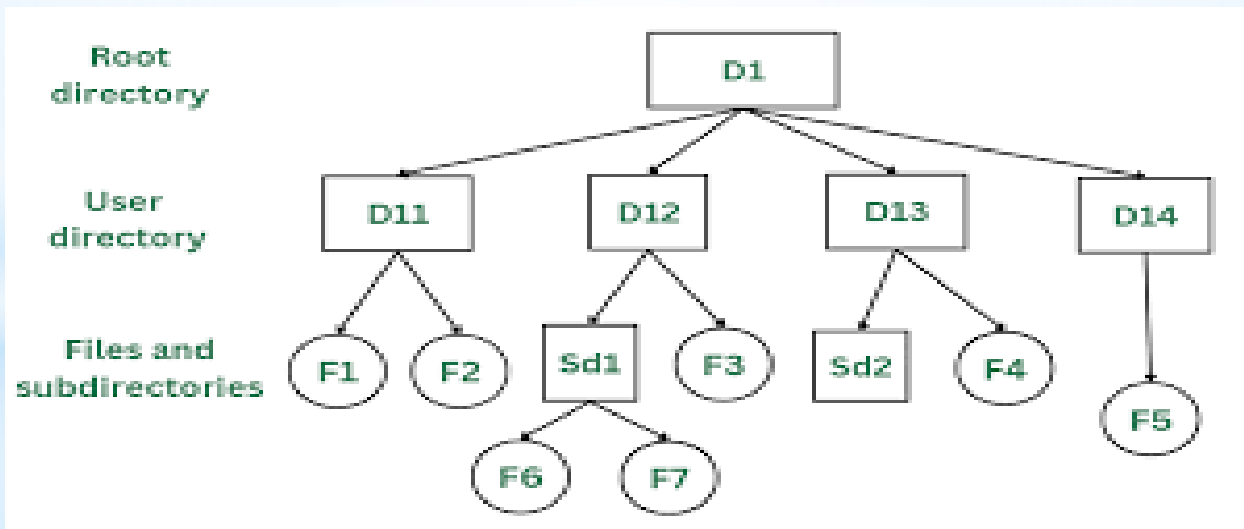
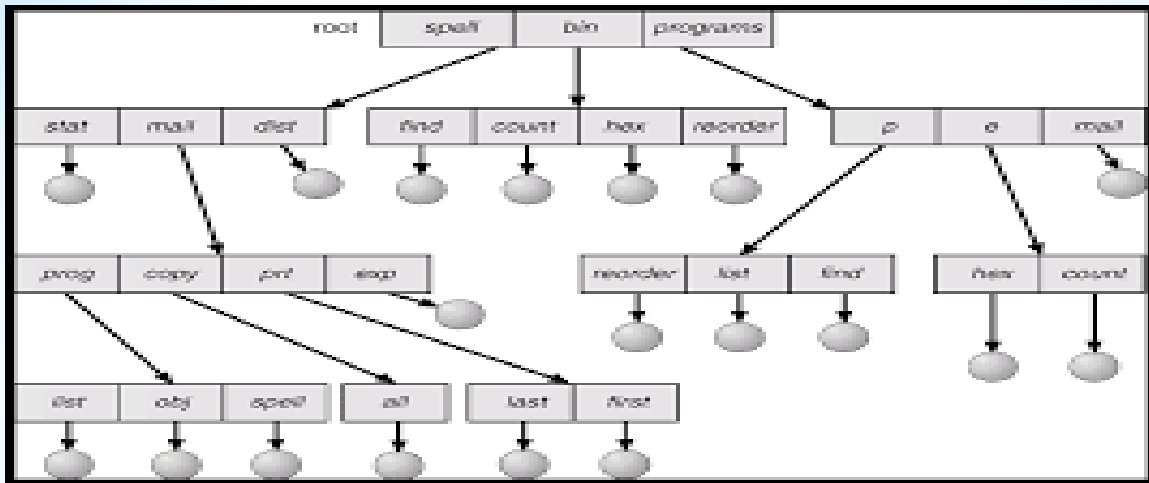
2) Two Level directory

- * In two level directory systems, we can create a separate directory for each user.
- * There is one master directory which contains separate directories dedicated to each user. For each user, there is a different directory present at the second level, containing group of user's file.
- * The system doesn't let a user to enter in the other user's directory without permission.



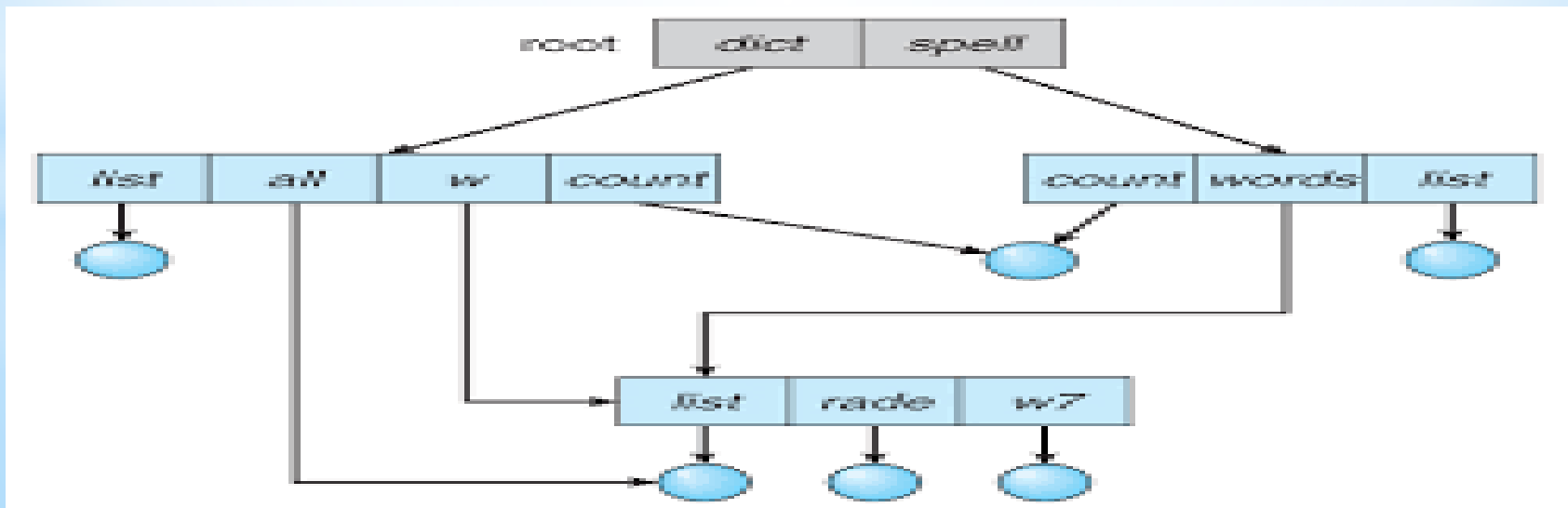
3) Tree Level directory

- * Tree structured directory system, any directory entry can either be a file or sub directory. Tree structured directory system overcomes the drawbacks of two level directory system. The similar kind of files can now be grouped in one directory.
- * Each user has its own directory and it cannot enter in the other user's directory. However, the user has the permission to read the root's data but he cannot write or modify this. Only administrator of the system has the complete access of root directory.
- * Searching is more efficient in this directory structure. The concept of current working directory is used. A file can be accessed by **two types** of path, either **relative or absolute**.
- * Absolute path name begins at the root and follows a path down to the specified file.
- * Relative path name defines path name from the current directory.



4) Acyclic Graph Directory

- * The tree structured directory system doesn't allow the same file to exist in multiple directories therefore sharing is major concern in tree structured directory system.
- * We can provide sharing by making the directory an acyclic graph. In this system, two or more directory entry can point to the same file or sub directory. That file or sub directory is shared between the two directory entries.
- * These kinds of directory graphs can be made using links or aliases. We can have multiple paths for a same file. Links can either be symbolic (logical) or hard link (physical).



5) General Graph Directory

- * One's serious problem with using an acyclic graph structure in ensuring that there are no cycles.
- * If we start with a two level directory and allow users to create subdirectories.
- * It should be easy to see that simply adding new files and subdirectories to an existing tree structured directory preserves the tree structured nature.
- * When we add links to an existing tree structured directory the tree is destroyed resulting in a simple graph structure.

